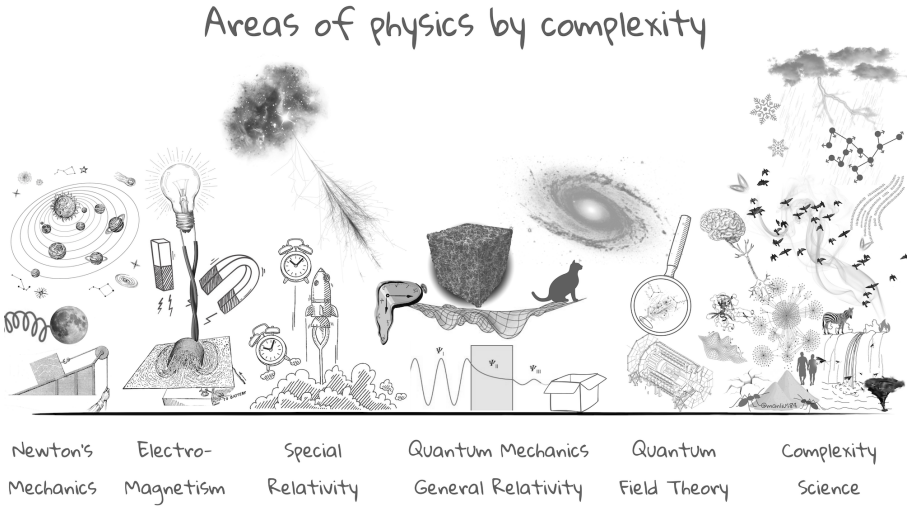
Final Report

Physics of Complex Networks: Structure and Dynamics



Università degli Studi di Padova



Project # 45: Network reconstruction

Project # 24: Entanglement percolati

Méndez Rodríguez de Tembleque, Álvaro

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EvoSim simulations and cleaned trajectory data

Task leader(s): Méndez Rodríguez de Tembleque, Álvaro

Project 45 (score 1.2). Objective 1: selection or generation of EvoSim simulation outputs.

The simulator and what it emits

Scenarios

Cleaning and harmonisation

Data availability

Reconstructed networks and temporal statistics

Task leader(s): Méndez Rodríguez de Tembleque, Álvaro

Project 45. Objectives 2–3: at least two network representations from the same trajectories, and their temporal statistics.

Two network definitions

Lineage network L_t

Metabolic co-activity network M_t

Summary of network types

Snapshot file schema

Snapshot visualisation

[figure placeholder — `figures/p45/snapshots.pdf`, produced by `code/02_networks.R`]

Figure 1: Lineage (L_t) and metabolic co-activity (M_t) snapshots for run `rich_punctuated_10x`, before and after the collapse at epoch 205.

Temporal statistics

[figure placeholder — `figures/p45/temporal_stats.pdf`, produced by `code/03_stats.R`]

Figure 2: Temporal network statistics for L_t and M_t over run `rich_punctuated`.

Network structure and eco-evolutionary dynamics

Task leader(s): Méndez Rodríguez de Tembleque, Álvaro

*Project 45. Objectives 4–5: do the reconstructed networks track population-level quantities, and does structure change **before** a transition?*

Population-level observables

Joint dynamics

[figure placeholder — `figures/p45/overlay.pdf`, produced by `code/04_coupling.R`]

Figure 3: Network descriptors overlaid on population-level observables, run `rich_punctuated`. Dashed lines mark detected transitions.

Quantitative coupling

Structure before transitions

Scenario comparison and interpretation

Task leader(s): Méndez Rodríguez de Tembleque, Álvaro

Project 45. Objectives 6–7: comparison across simulation scenarios, and which representation is most informative.

Across scenarios

Which representation is most informative

Optional extensions

Limitations

Entanglement percolation: model and strategies

Task leader(s): Méndez Rodríguez de Tembleque, Álvaro

Project 24 (score 0.6). Reproduction of Cuquet and Calsamiglia, "Entanglement Percolation in Quantum Complex Networks", PRL 103, 240503 (2009) (Cuquet and Calsamiglia 2009).

Setting

$$p \;=\; \min[1,\, 2(1-\lambda_0)]\,, \qquad p_2 \;=\; \begin{cases} 2p - p^2, & \text{without distillation,} \\ \min\!\big(1,\, 2p - p^2/2\big)\,, & \text{with distillation.} \end{cases} \tag{1}$$

The q-swap

[figure placeholder — `figures/p24/qswap.pdf`, schematic redrawn from the paper]

Figure 4: Entanglement swapping (left) and the 5-swap transformation (right): a q-star centred on the measured node becomes a q-cycle, and the central node is detached. Redrawn after (Cuquet and Calsamiglia 2009).

What is reproduced here

Generating-function theory of q-swap percolation

Task leader(s): Méndez Rodríguez de Tembleque, Álvaro

Project 24. Analytic treatment of random graphs with arbitrary uncorrelated degree distribution.

Classical baseline

$$G_0(x) = \sum_{k\geq 0} P(k) x^k, \qquad G_1(x) = \frac{G_0'(x)}{G_0'(1)}, \tag{2}$$

$$H_1(x) = \big(1 - p_2\big) + p_2\, x\, G_1\big(H_1(x)\big), \qquad H_0(x) = x\, G_0\big(H_1(x)\big). \tag{3}$$

After the q-swap

$$\tilde{H}_1(x) = 1 - p_2 + p_2\, x\, G_1\big(\tilde{H}_1(x)\big) + \sum_{q\geq 2} \gamma_q\, \tilde{H}_{1,q}(x). \tag{4}$$

$$\tilde{H}_0(x) = x\, G_0\big(\tilde{H}_1(x)\big) + x \sum_{q\geq 2} \gamma_q \beta_q P(q) \Big\{ 1 - \big[\tilde{H}_1(x)\big]^q \Big\}, \qquad \tilde{S} = 1 - \tilde{H}_0(1). \tag{5}$$

Solving in practice

Bethe lattice sanity check

Erdős–Rényi and scale-free networks

Task leader(s): Méndez Rodríguez de Tembleque, Álvaro

Project 24. Analytic results checked against Monte Carlo simulation.

Erdős–Rényi

Scale-free with exponential cutoff

Numerical validation

[figure placeholder — `figures/p24/er_sf.pdf`, produced by `code/02_simulate.R`]

(a) Gain Δ in the ER model versus mean degree z for the 2-swap, $\{2,3\}$-swap, $\{2,3,4\}$-swap and optimal strategies.

(b) GCC size versus p for $z = 2.5$ without distillation, before and after the $\{2,3\}$-swap; points are simulations, lines are

1. (c) As (a) for the scale-free model with $\gamma = 1$, versus the cutoff κ. (d) GCC size for a real WWW sample with degree cutoff $k = 15$. Compare with Fig. 2 of (Cuquet and Calsamiglia 2009).

Figure 5

Small-world networks and discussion

Task leader(s): Méndez Rodríguez de Tembleque, Álvaro

Project 24. A case the generating-function theory cannot reach.

Why small-world needs simulation

[figure placeholder — `figures/p24/sw.pdf`, produced by `code/02_simulate.R`]

Figure 6: Small-world model with 2-swap and distillation. (a) p_c versus the shortcut density ϕ before and after the operation; the solid line is the classical analytic solution. (b) GCC size versus p at $\phi = 0.25$. Compare with Fig. 3 of (Cuquet and Calsamiglia 2009).

Discussion

References

Cuquet, Martí, and John Calsamiglia. 2009. "Entanglement Percolation in Quantum Complex Networks." *Physical Review Letters* 103 (24): 240503. https://doi.org/10.1103/PhysRevLett.103.240503.

Appendix: Project 45

Run inventory and integrity checks

Network file conventions

Additional statistics

Optional extensions

Appendix: Project 24

Derivation details

Numerical methods

Supplementary figures